

```

#include <stdio.h>

#include <stdlib.h>

typedef struct Node {

int data;

struct Node* prev;

struct Node* next;

} Node;

Node* createNode(int value) {

Node* newNode = (Node*)malloc(sizeof(Node));

newNode->data = value;

newNode->prev=NULL;

newNode->next=NULL;

return newNode;

}

void displayList(Node* head) {

Node* temp = head;

printf("\nDoubly Linked List Structure:\n");

printf("| %-10s | %-10s | %-5s | %-10s |\n", "Node Addr", "Prev Addr", "Data", "Next Addr");

while (temp != NULL) {

printf("| %-10p | %-10p | %-5d | %-10p |\n",

(void*)temp,

(void*)temp->prev,

temp->data,

(void*)temp->next);

temp = temp->next;

}

}

int main() {

Node* N1 = createNode(1);

Node* N2 = createNode(2);

Node* N3 = createNode(3);

```

```
Node* N4 = createNode(4);
```

```
Node* N5 = createNode(5);
```

```
N1->next = N2;
```

```
N2->prev = N1;
```

```
N2->next = N3;
```

```
N3->prev = N2;
```

```
N3->next = N4;
```

```
N4->prev = N3;
```

```
N4->next = N5;
```

```
N5->prev = N4;
```

```
displayList(N1);
```

```
free(N1);
```

```
free(N2);
```

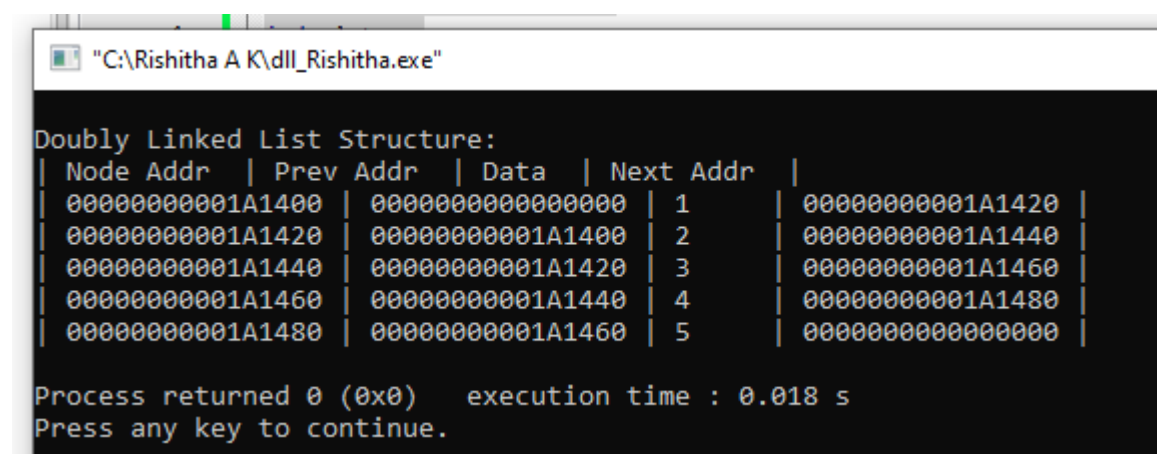
```
free(N3);
```

```
free(N4);
```

```
free(N5);
```

```
return 0;
```

```
}
```



```
"C:\Rishitha A K\dll_Rishitha.exe"

Doubly Linked List Structure:
| Node Addr | Prev Addr | Data | Next Addr |
| 00000000001A1400 | 0000000000000000 | 1 | 00000000001A1420 |
| 00000000001A1420 | 00000000001A1400 | 2 | 00000000001A1440 |
| 00000000001A1440 | 00000000001A1420 | 3 | 00000000001A1460 |
| 00000000001A1460 | 00000000001A1440 | 4 | 00000000001A1480 |
| 00000000001A1480 | 00000000001A1460 | 5 | 0000000000000000 |

Process returned 0 (0x0)   execution time : 0.018 s
Press any key to continue.
```